Smart Local Bus Route Finder

A Database-Driven Analytical Dashboard for Karachi's Public Bus Network

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Database Systems Final Project · SZABIST University, Karachi

THE PROBLEM

Why this project

- Millions ride Karachi's buses daily, but there is no single system telling them which bus to take.
- Passengers do not know transfer points, fares, or travel time in advance.
- Operators have no consolidated view of ridership, revenue, reliability, or feedback.
- Route information is scattered across informal sources instead of one database.

THE SOLUTION

A three-tier web application

- **Database:** PostgreSQL on Neon, 21 related tables
- **Backend:** Node.js + Express REST API (JSON)
- **Frontend:** HTML, CSS, Bootstrap, JavaScript
- **Charts:** Chart.js on a single dashboard page
- **Map:** Leaflet + OpenStreetMap route view

21
database tables

20+
REST API endpoints

13
analytical charts

TECHNOLOGY STACK

Built with

PostgreSQL 16 · Neon (cloud DB) · Node.js · Express.js · node-postgres · HTML5 · CSS3 · Bootstrap 5 · JavaScript (Fetch / AJAX) · Chart.js · Leaflet · Vercel · Git / GitHub

Data coverage

- BRT: Green Line, Red Line, Orange Line
- Feeder: Peoples Bus Service R-10, R-12
- Electric: EV-1, EV-3 · shared interchange stops enable transfers

DATABASE DESIGN

21 tables, fully constrained

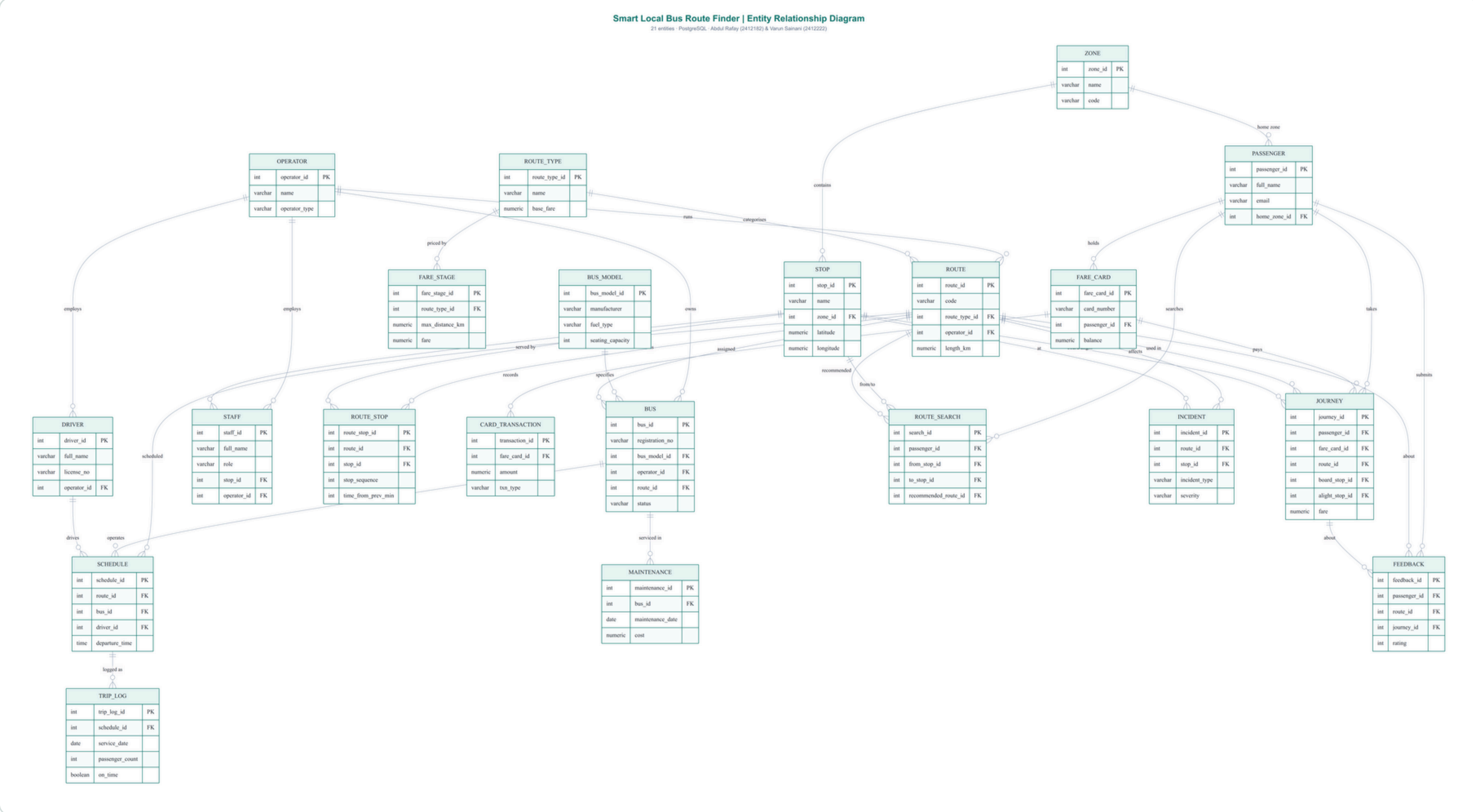
- **Network:** zone, operator, route_type, route, stop, route_stop, fare_stage
- **Fleet and ops:** bus_model, bus, driver, staff, schedule, trip_log, maintenance
- **Passengers and revenue:** passenger, fare_card, card_transaction, journey, route_search, feedback, incident

22,000+
sample rows seeded

6,000
passenger journeys

PK / FK
+ check constraints throughout

ENTITY RELATIONSHIP DIAGRAM



REST API

Endpoints return JSON

Group	Examples
Analytics (13)	`/api/analytics/summary, ridership-by-route, peak-hours, on-time-performance`
Route finder	`/api/finder/stops, /api/finder/plan?from=&to=`
CRUD (8 resources)	`GET, POST, PUT, DELETE on routes, stops, passengers, ...`

- Connects to PostgreSQL through a pooled connection
- Performs CRUD and fetches analytical / statistical data

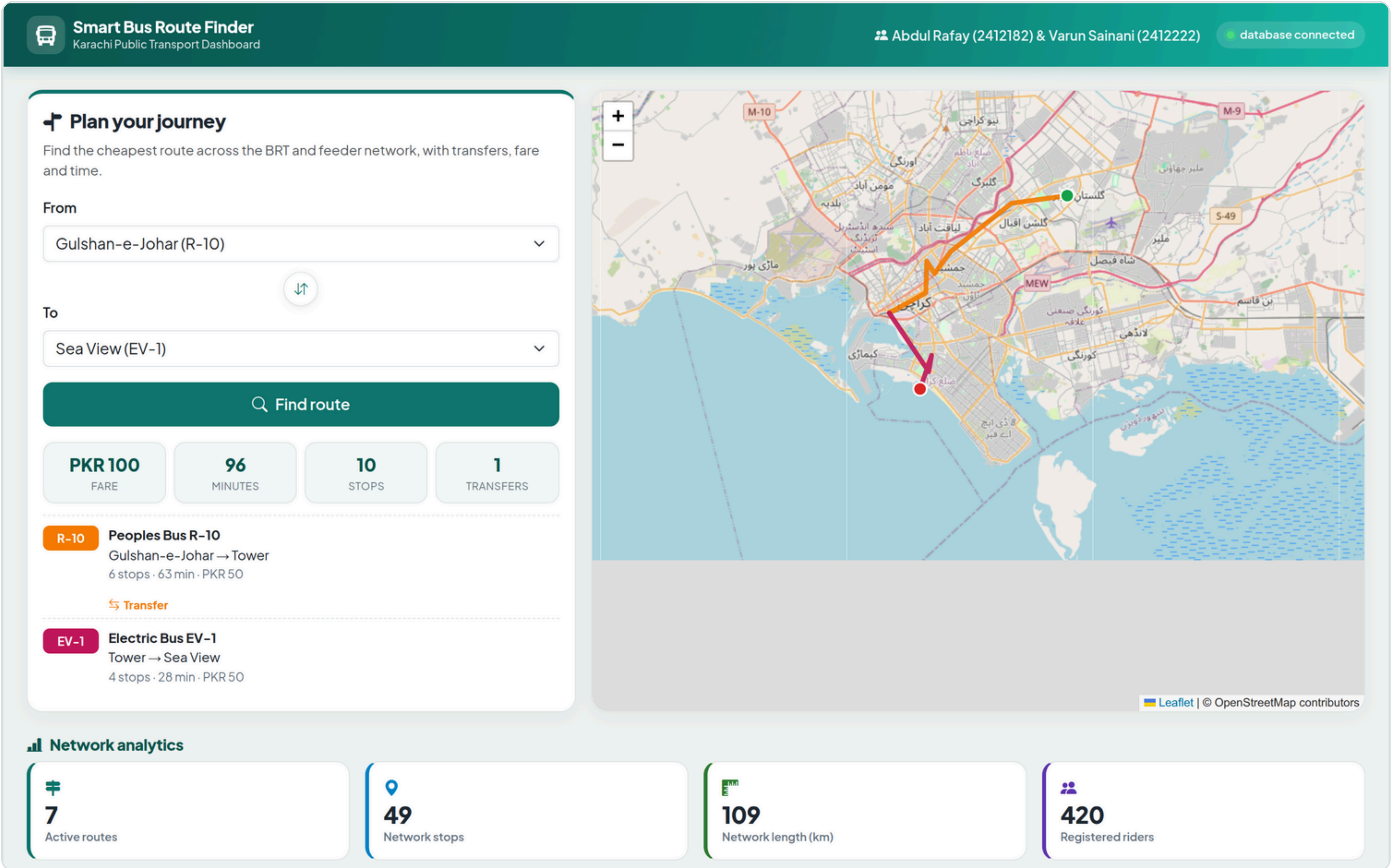
Dijkstra shortest path

- The network is a graph of `(route, stop)` states.
- Riding moves along a route; a transfer switches routes at a shared stop with a time penalty.
- Dijkstra's algorithm finds the lowest total travel time.
- Fare for each leg is computed from the fare-stage table; results show legs, fare, time, stops, and transfers.

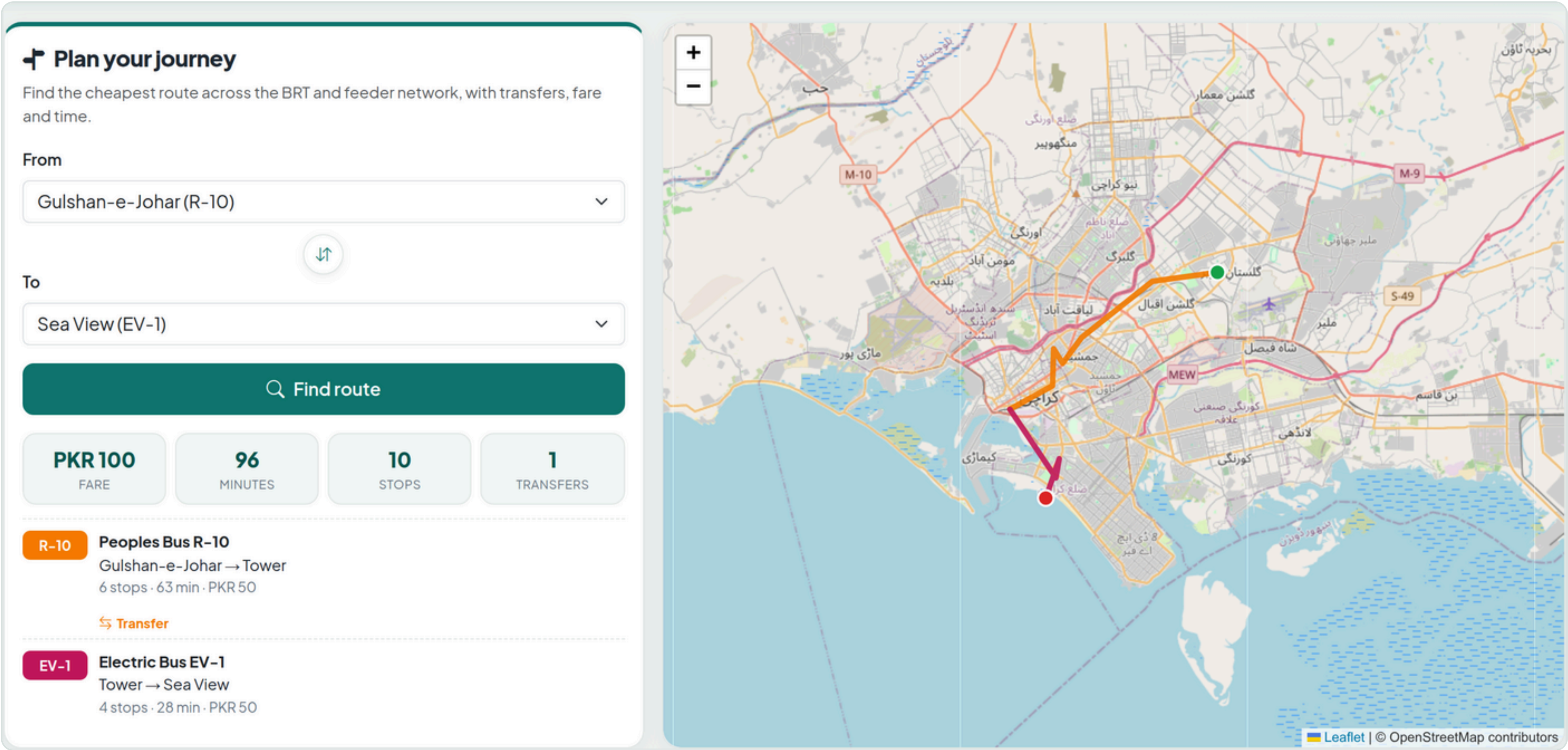
Gulshan-e-Johar → Sea View

R-10 to Tower, transfer to EV-1: PKR 100, 96 min, 1 transfer

ANALYTICAL DASHBOARD



ROUTE FINDER AND MAP



TESTING

Verified end to end

Test	Result
Health and database connectivity	Passed (200)
13 analytical endpoints	Passed (valid JSON)
CRUD round trip (POST, GET, PUT, DELETE)	Passed (201, 200, 200, 200)
Route finder with transfer	Passed (correct fare and time)
Invalid input handling	Passed (400 / 404)

Thank you

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